SEX-LINKED GENETICS

In humans the first 22 chromosomes are called ***autosomal chromosomes*** and are responsible for many of the inherited traits such as eye colour, skin pigment and hair colour.

The 23rd pair of chromosomes, the sex chromosomes, look different and carry different genes. The sex chromosomes are not necessarily identical in their appearance and gene content.

The X chromosome carries genes that affect body structures and functions. These characteristics are called ***X-linked*** because in most cases there is no corresponding allele on the Y chromosome. Genes carried only on the Y chromosome are called ***Y-linked***.

Generally, the characteristics carried on X or Y chromosome are called ***sex-linked***.

Colour blindness is a clear example of X-linked inheritance. Normal colour vision is determined by the presence of the dominant gene, which can be called B rather than C so that it is not mixed up with a lower case c. Colour blindness results from the recessive gene b.

e.g. A female, with her two X chromosomes, can be either homozygous X^BX^B (called ***normal***) or heterozygous X^BX^b (called a ***carrier***) and still have normal colour vision.

She will be colour blind only if she has two recessive alleles, X^bX^b.

A male with only one X chromosome displays the trait that the chromosome carries.

i.e. X^BY is normal. X^bY is colour blind.

There are a number of conditions that are X-linked, including types of ***haemophilia***, ***diabetes*** and ***muscular dystrophy***.

The table below can be used to determine if a characteristic is X-linked recessive or dominant in a pedigree.

X-linked recessive inheritance

- Most affected individuals are male.
- Affected males result from mothers who are affected or who are known to be carriers (heterozygotes), and have affected brothers, fathers or maternal uncles.
- Affected females come from affected fathers and affected or carrier mothers.
- The sons of affected mothers should be affected.
- Approximately half the sons of carrier females should be affected.
- Examples include haemophilia and Duchenne muscular dystrophy.

X-linked dominant inheritance

- Traits do not skip generations.
- Affected sons must come from affected mothers.
- Approximately half of the children of an affected heterozygous female are affected.
- Affected females come from affected mothers or fathers.
- All the daughters, but none of the sons, of an affected father are affected.
- Very few conditions show sex-linked dominant inheritance; Rett syndrome is one example.

Example

A normal male and a heterozygous (carrier) mother for colour blindness have children together. What is the probability that a boy born to the couple will have colour blindness?

X^BY x X^BX^b

♂ ♀

♀ \ ♂	X^B	Y
X^B	$X^B X^B$	$X^B Y$
X^b	$X^B X^b$	$X^b Y$

F1	**genotypes**	$X^B X^B$	$X^B X^b$	$X^B Y$	$X^b Y$
	phenotypes	normal female	carrier female	normal male	colour blind male

<u>Hence 50 % of the males will be colour blind.</u>